



Algorithmic Redistricting of Irish Election Boundaries: A Framework for Rethinking Dáil Constituencies

Theoretical Physics Student Association

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1 Introduction

Proportional Representation and the need for Electoral Redistricting

An electoral district, or constituency, is a geographical subdivision of a state created to provide its population with representation in the larger state's legislative body[(1)]. Electoral redistricting is the process of redrawing the electoral boundaries that make up constituencies. At its most basic level, the principle motivating the necessity of redistricting is that of proportional representation. Proportional representation (PR) is an electoral system in which the underlying distribution of electorate support for each political party is reflected proportionately in the elected representative body. PR is a core feature of the Irish democratic landscape. To accommodate PR, constituencies geographically divide the state such that the representatives elected to each constituency are individually responsible for the same number of electorate as any other representative in the state (although this will technically fall short on the principle of PR for subtle reasons explained later). Between election cycles, as the population density across the state alters, so too must the geographical boundaries of the constituencies. Election redistricting is a necessary consequence of PR.

Electoral Redistricting in Ireland

This paper is written within the Irish context, so we must clearly define the following basic terms for future reference:

- The national parliament of Ireland is referred to as Dáil Éireann, or simply the Dáil.
- An elected representative sitting in the Dáil is referred to as a Teachta Dála (TD).
- Ireland is divided up into 3,440 regions known as electoral divisions (EDs).
- These EDs are currently split between 39 constituencies. The current configuration of constituencies in Ireland is shown in Figure 1.1. The constituency boundaries can be changed by transferring EDs from one constituency to another. The number of constituencies is not fixed and may be altered in a redrawing of boundaries.
- Ireland is also divided into 26 counties, which are administrative regions governed by bodies called county councils. The location of constituency boundaries is independent of the position of county boundaries, but it is considered desirable for constituencies to conform to county boundaries as far as is practicable.

On 9 February 2023, a new state body called the Electoral Commission was established to oversee elections in Ireland. One of the roles of the Electoral Commission is reviewing the Dáil Éireann constituencies, and making a report and recommendations in relation to possible changes to constituency boundaries. When deciding on any changes to the current constituency boundaries, the Electoral Commission is required to observe the following provisions of the Constitution [2, 3]:

- Article 16.2.2° of the Constitution provides that: *The number of members shall from time to time be fixed by law, but the total number of members of Dáil Éireann shall not be fixed at less than one member for each thirty thousand of the population, or at more than one member for each twenty thousand of the population.*
- Article 16.2.3° of the Constitution provides that: *The ratio between the number of members to be elected at any time for each constituency and the population of each constituency,*

as ascertained at the last preceding census, shall, so far as it is practicable, be the same throughout the country.

There is significance jointly given to other objectives which are detailed Section 57 of the Electoral Reform Act 2022. Stated briefly, these objectives include:

1. There must be between 171 and 181 (inclusive) elected Dáil representatives.
2. Each constituency will have 3, 4, or 5 members (seats).
3. County boundaries should be preserved.
4. Constituencies must be contiguous (there cannot be isolated regions).
5. Geographic considerations to do with physical features and population density should be considered.
6. Continuity of the current arrangement of constituencies should be upheld as much as possible.

There is no mention of the relative importance of the themes of county boundaries, geographic consideration and constituency continuity, though it is generally understood that PR trumps these themes. When searching for satisfactory configurations, terms (i), (ii), and (iv) cannot be broken, whereas the remaining (iii), (v), and (vi) can and will be transgressed in the final boundaries to some extent.

Algorithmic Redistricting in Ireland

To the author’s knowledge, an algorithmic redistricting aid is not presently employed. Fortunately, the manner of redistricting in Ireland, that of moving EDs between constituencies, lends itself very well to a mathematical representation. As a result, the time-intensive process of manually generating and evaluating proposed constituencies can be dispensed with by using an appropriate algorithm.

The main objective of this paper is to outline algorithmic techniques that can tackle the redistricting mathematically, such that the space of possible solutions can be searched in a unbiased and efficient manner (save for bias in the numerical-stochastic sense, though this is a rather academic and irrelevant point for our purposes). Later in the text, we briefly touch on further-reaching use-cases for such an algorithmic approach, namely in assessing existing legal rules in achieving fair representation, and in testing the resilience of the rules against gerrymandering.

Technical Problem Overview

We will refer to any one composition of Dáil constituencies and their boundaries as a “configuration”. From a mathematical point of view, the number of possible configurations is truly enormous. The vast majority of the space of all possible configurations are completely unreasonable from a legal perspective; a configuration selected at random will likely dramatically fail the 171-181 seat restriction, the 3-5 seat range, the county boundary, and the contiguity requirements simultaneously. Only a tiny sliver of the space of all possible configurations is satisfactory according to the requirements; and even within all valid configurations, there still exists a hierarchy from passable to optimal configurations that is impossible to rigorously probe by a human expert.

Automating the task roughly splits into two main axes; mathematically quantifying the legal rules, and, having then established a means to differentiate good from bad configurations, efficiently navigate the possibilities in a manner which systematically finds the better solutions. It should be noted that this process is not perfect with respect to either axis. In the case of codifying the rules, there is the ambiguity of legal interpretation, layered on with further technical difficulty in faithfully translating the relative importance of the legal rules to one another into a multi-objective mathematical summary that has the desired effect. In the case of the search, there is the ambiguity of different configurations admitting the same score under the chosen objective function. Both of these issues are addressed to some extent later in the text.

Very roughly speaking, and at its most basic level, the algorithms execute a search of possible configurations as follows: Consider some initial configuration. Quantify how closely the configuration adheres to the rules. Make a change, e.g., take an ED, and make it part of a different constituency. Re-evaluate how closely the changed configuration adheres to the rules. Use the difference in the performance of the changed, and unchanged configuration to inform the search. Continue this process iteratively until the calculated adherence to the rules is satisfactory. If the rules have been parsed well into maths, and the search is effective, the final configuration will be a set of constituencies that respect the Constitution and Terms of Reference.

The algorithmic solution must be formulated in a way that is interpretable and open to scrutiny. This translates to using a minimal number of tunable parameters, such that the value of each parameter can be subject to legal debate. A “black-box” solution, which would include Deep Learning approaches that invoke thousands of parameters which admit highly non-linear inter-dependencies, is deliberately avoided for this reason.

1.1 Context: Historical Context

- Boundary commissions -> now we have Election commissions

- MATTHEW WILL WRITE - WILL HIGHLIGHT IN OTHER SECTION HOW THEY HAVE DEALT WITH CERTAIN RULES

We would like to highlight that in most cases (the ones that are well known). Computational methods have been deployed to get a better democracy due how electoral boundaries are drawn by entities that are not 100% independent. This is not the case in Ireland. The problem being faced here is that of how can we improve and refine the already functional system. We believe given the establishment of Ireland’s new independent electoral commission, An Coimisiún Toghcháin. That there is now a opportunity for introduce these sort of computational methods. We believe that their introduction could help achieve and enhance the following

- Improve transparency: Auditable trail for decisions
- Improve turn around times. Speeding up the process of finding valid maps
- Improve efficiency and provide decision support to our new electoral commission who will faced with a significant challenge in the coming years when they will need to review and potentially redraw the following electoral boundaries: EU, LEA and Dáil

MOST RECENT CENSUS Introduction of a new way of defining urban areas. They utilised algorithms to assist in this process, but still deferred to human in the loop to solve issues in particular cases

Make the point that this is a president (precedent?) for utilising algorithms to help define areas that are valid under a rule set that is feed into the algorithm

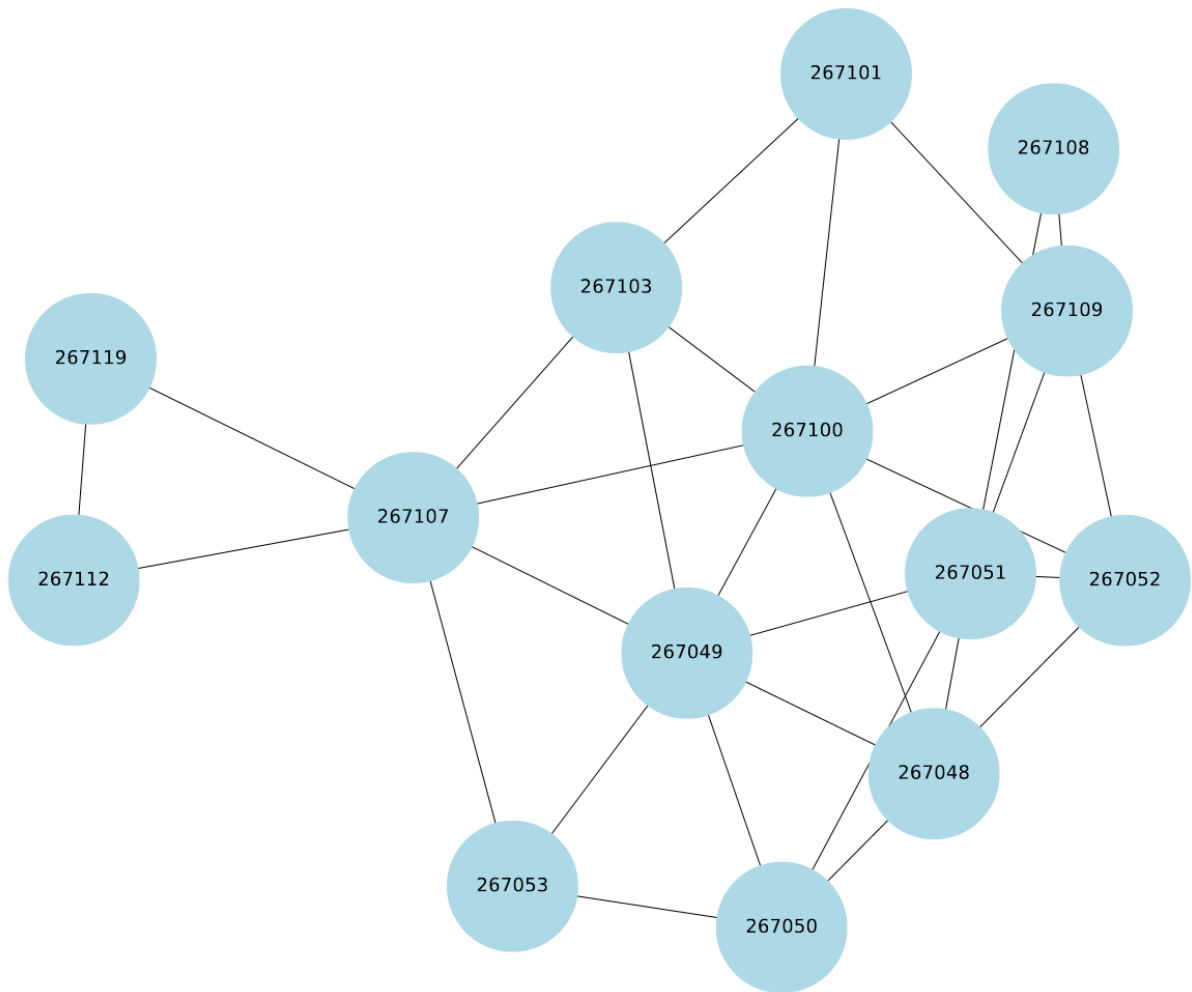


Figure 1: The numbers originate from the Electoral District ID in the CSO dataset (A “pandas GeoDataFrame” file). The lines connecting EDs to one another indicate if they are land connected. There are exceptions to this land-connected definition of EDs being neighbours or not; these will be explored in the section on contiguity. This may be viewed as a mathematical “graph”, where the circles signifying EDs are “nodes”, and the lines connected EDs are “edges”.

2 Challenges (Michael)

A function that characterises error is a requisite to a process of optimisation. We will now explore the framing of the individual legal objectives as optimisable sub-components of such a function.

2.1 Incorporating Geographical Information

Before exploring the framing of the legal objectives as optimisable functions, we must first consider how ED geographical information is held within the algorithm. There is no set way to represent the information, but certainly all forms must include neighbour information, roughly in line with Fig. [1].

The optimisation process will rely on this representation of the geography to identify which EDs are reasonable to add to a given constituency during the search. A geographically informed search, wherein EDs do not change their constituency membership to a random constituency,

but to a nearby/neighbouring constituency, is much significantly faster than a random search. Other geographical information to do with constituency and community boundaries is overlaid on this neighbourhood information; this is explained later.

2.2 Proportional Representation and “Variance”

In line with Article 16.2.3° of the Constitution, let us base our discussion of quantifying voter representation on the National Ratio, defined as follows;

$$\text{National Ratio} = \frac{\text{Recorded Census Population}}{\text{Number of Dáil seats}} \quad (1)$$

For a given hypothetical constituency, we will refer to the population of its electorate as the Proposed Population. In a previous work submitted to the Electoral Commission as a submission for the open consultation on the electoral boundaries, a measure known as Seat Equivalent Representation (SER) was introduced. It is defined as follows:

$$\text{Seat Equivalent Representation} = \frac{\text{Proposed Population}}{\text{National Ratio}} \quad (2)$$

This measure directly captures the number of seats a hypothetical constituency deserves. We can see this by analysing the units of the measure. The National Ratio is measured as

$$[\text{National Ratio}] = \frac{[\text{People}]}{[\text{Seats}]} \quad (3)$$

With this in mind, now consider the units of the Seat Equivalent Representation metric;

$$[\text{Seat Equivalent Representation}] = \frac{[\text{Proposed Population}]}{[\text{National Ratio}]} = \frac{[\text{People}]}{\frac{[\text{People}]}{[\text{Seats}]}} = [\text{Seats}] \quad (4)$$

The Seat Equivalent Representation tells us how many TDs should be assigned to a proposed constituency, given the population of said constituency. The nice thing about this metric is unlike population number, it centres representation as measured in seats as the key concept being addressed. We can always switch back to the population via the following formula,

$$\text{Proposed Population} = \text{Seat Equivalent Representation} \times \text{National Ratio} \quad (5)$$

In reality, constituencies will often have a non-integer SER, meaning that when an integer number of TDs are assigned to the constituency, the electorate will often be marginally over- or under-represented. The higher the discrepancy between the SER and the assigned number of seats, the greater the degree of misrepresentation. The previous Constituency Commission reports have defined “Variance from the National Average” based on population to c, which is 1-to-1 equivalent to the same definition in terms of SER:

$$\text{Variance from the National Average} = \frac{\text{Seat Equivalent Representation} - \text{Assigned Seats}}{\text{Assigned Seats}} \quad (6)$$

A positive value of 0.05 indicates a variance of 5% above the National Average Representation, while a negative value of -0.05% indicates a variance of -5% below the National Average. In a recent submission to the electoral boundary consultation, our team submitted an alternative

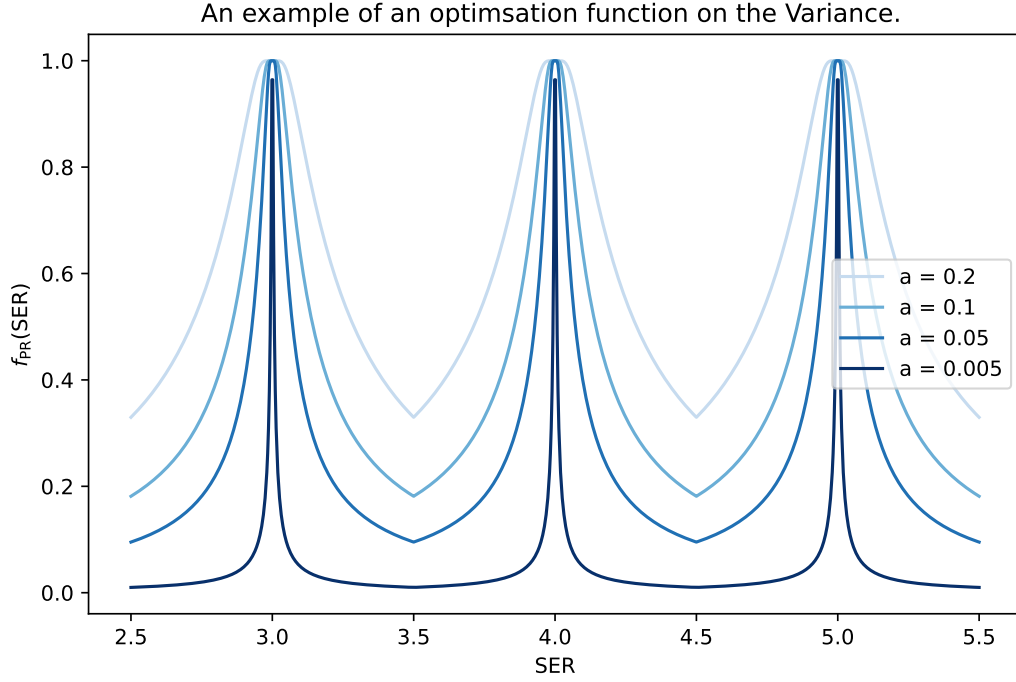


Figure 2: Functions suitable for optimising over SER all bear common points of extremal values at integer values of SER. The functions are constructed to yield the least favourable result at midway points between the integer SER values, so that the search process tends away from non-representative configurations.

definition of variance which is provably a more democratic representation of variance from the national average. For more information, see the appendix.

Optimisation tasks work by iteratively altering some mathematical circumstance so that a function acting on said circumstance may take on an extremal value. For our purposes, whatever function the SER is considered within, it is crucial that said function’s extremal values occurs when the SER values are either 3, 4, or 5, as dictated by the terms of reference. As a illustrative example of such a function, consider:

$$f_{\text{PR}}(\text{SER}) = 1 - e^{-\frac{a}{(\text{Assigned Seats}) \times |\text{Variance}|}}, \quad (7)$$

Altogether, this optimisation function yields a class of surfaces as in Fig. 2. During the optimisation process, departures from integer SER is penalised. Of course, this then shifts a great deal of responsibility for adhering to proportional representation onto the definition of variance to begin with, as the optimisation function is constructed about that.

2.3 Contiguity

The contiguity of a constituency refers to whether the constituency is whole, or split into different regions that are not connected by land. Ignoring these exceptions momentarily for the sake of clarity, there are standard computational techniques that can calculate whether a region is contiguous or not stemming from graph theory. In the Irish context, contiguity is a hard requirement. From an optimisation perspective, there is a design choice; do we constrain our search to only configurations with contiguous regions, narrowing and hastening the search, or do

we entertain non-contiguous configurations during the search, in the hopes that by the time the search completes, might it happen on a better configuration (In optimisation speak, we refer to this as “avoiding local minima”)? If we constrain the search to only contiguous configurations, this rules out being able to begin the search with a completely unbiased, randomised starting point, since the constituencies would be very scattered at the start. In order to remain efficient however, some amount of search narrowing is required. The resolution is to include a term that severely punishes breaking contiguity, so that it is heavily disincentivised, but to also restrict the EDs eligible to change constituency membership to those at constituency boundaries, and to ensure that they must flip membership to that of a nearby constituency, rather than any random one. This will have the effect of constituencies morphing, growing, and shrinking during the search, rather than peppering the map in a non-contiguous way, while still allowing islands to temporarily form if all other objectives deem it favourable to do so.

There are some exceptions to the land-connected contiguity definition, such as where electoral districts which are intrinsically split. These examples are pictured in Fig. [3]. However, when the entire ED is absorbed into the nodes as in Fig. [1], the algorithm treats these as being land-connected, which is a fine approximation for our purposes.

2.4 Compactness

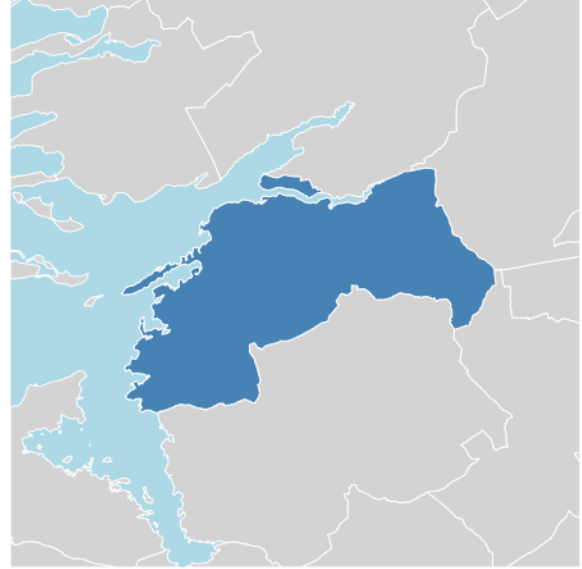
The word gerrymander comes from the combination of “Gerry”, after Elbridge Gerry, and “Salamander”, alluding to the meandering shape of a deliberately manipulated constituency. To avoid these highly-unnaturally shaped constituencies, a compactness optimisation is performed. As was the case with SER and variance, in order to optimise for it, we first need to measure it, and there are a number of ways to perform this. The most intuitive, but also most computationally demanding measure of compactness is via a calculation of the polygon enclosing the constituencies, what is known as the “convex hull”. This is pictured in Fig. [4]. The computationally demanding aspect is having to continually reference the geometry after every edition is made. Alternatively, we may instead optimise to minimise the number of external boundary ED neighbours, or maximise the number of internal boundary ED neighbours, both of which will induce roughly circular (or at the very least non-serpentine) constituencies.

2.5 County Boundaries and Geographic Considerations

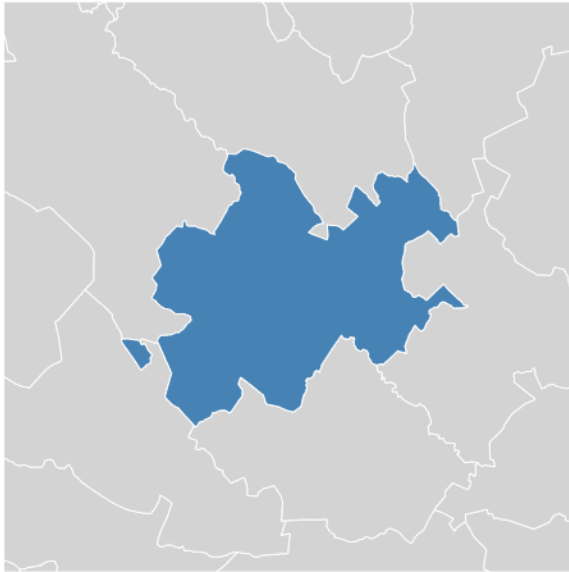
The proof-of-concept algorithms outlined in the paper have not been executed at a scale at which the effects of optimising against county boundaries can be observed. However, the general idea for constructing such a function is straightforward. If we consider a representation of the EDs in the form of Fig. [1], county boundaries are the collection of graph edges connecting EDs that are on either side of a county boundary. The most basic measure of boundary breaking would be to count the number of times a county boundary edge connects two EDs that are members of the same constituency. This count is then minimised during the search, minimising the county breaks. This measure may be somewhat at odds with the practice of the Electoral Commission, which considers a constituency that perfectly encompasses two counties to not be breaking county borders. This discrepancy is addressed in more detail in section 5.1. There are also certain cases where a community that is not very geographically localised may be adamantly against being split by an electoral boundary. These may be introduced into the algorithm in much the same way as the county boundaries; designating graph edges that will count towards a penalty in the event that the EDs connected by the edges are members of different constituencies.



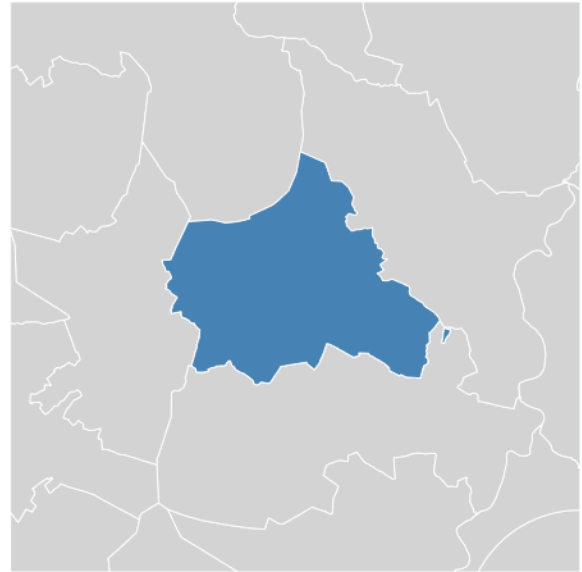
(a) Abbey, Clare



(b) Drumacoo, Galway East



(c) Moynalty, Meath East



(d) Downings, Kildare North

Figure 3: The four problematic non-contiguous boundary electoral divisions.

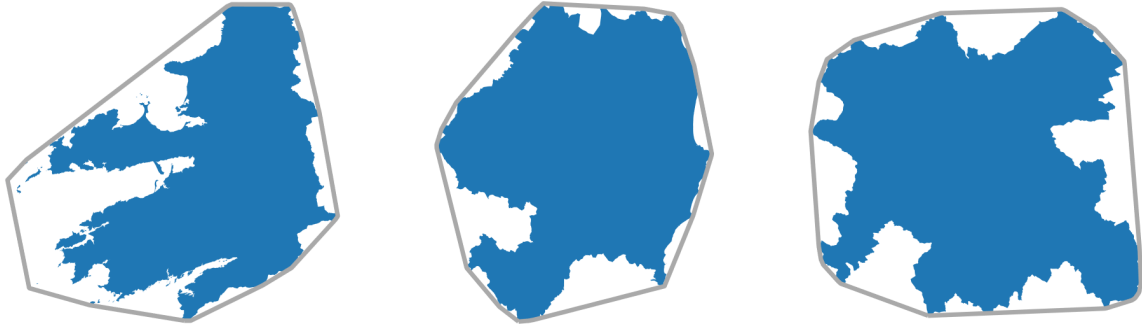


Figure 4: Caption

2.6 Temporal Boundaries

As outlined in the Terms of Reference, the Commission aims to maintain continuity in relation to the existing arrangement of constituencies. Arguably, there are two ways to ensure this in the algorithm; bias the search by using the existing boundaries as the starting configuration, optimise for it by penalising discontinuity, or use both approaches simultaneously. To optimise against discontinuity, we must first come up with a measure of the discontinuity. Let us introduce a basic function f_C for the purpose of illustrating this measure.

$$f_C(\text{Configuration}) = e^{-a_c n_{\text{people}} - b_c n_{\text{ED}}} \quad (8)$$

where n_{people} is the total number of people no longer in their original constituency, and n_{ED} being the number of EDs no longer in their original constituency. The constants a_c and b_c would change the relative weighting of each of these two considerations. The function f_C takes a maximum of 1 when $n_{\text{people}} = n_{\text{ED}} = 0$, so it is maximal when no change has happened, providing an extremal value to optimise towards. However, this approach is not complete, since it not sensitive to the underlying geography; f_C might yield the same penalty for a constituency with minor changes everywhere, and one with very drastic changes in one region. This geographic consideration is mathematically addressed in section 5.1. The following section 2.7 introduces a further complication.

2.7 Weighting Rule-Break by Population

Equation (8) introduced an interesting notion of having the breaking of the Term of Reference/Constitutional rule in question be contextualised by the number of people affected. In addition to continuity, this can also be applied mathematically to optimising SER variance (arguably a controversial move), county boundaries, and community boundaries. However, regarding the county, the Commission notes in the most recent Constituency Review Report;

“The transfer of a small population can add to the perception of lack of effective representation. A view was expressed in submissions that if a county boundary was to be breached, the breach should involve the transfer of a sufficiently large number of persons to ensure that the transferred population carried some weight.”

This is a difficult sentiment to measure and optimise around. A possible optimisation function on the SER lying over the county boundary is given in 5. The range of this figure is from zero to

one, since at one it is equally as favourable for a constituency in the county being encroached on to absorb the over-reach. The rapid drop-off at 0 reflects the Commission’s observation that a small population transfer can leave the transferred population feeling under-represented. The bump in the graph at an SER of 0.4 reflects the notion that transferring a larger number of people is preferable so that their vote carries weight. Facilitating the Commission’s sentiment that transferred populations should be substantial will introduce another three tunable parameters to control the position of the bump, namely the height (quantifying how much worse it is to transfer population as a block than to stay within), the SER value (how much of a population is acceptable), and the width (tolerance).

One quite technical point worth noting is that if the acceptable SER value is 0.5, this may introduce an issue in converging the search, since a symmetry has been introduced. If I have 2 constituencies, A and B, both with an SER of 4.5 to begin with, the situations where $A = 4$, $B = 5$ with B over-reaching is identical to $A = 5$, $B = 4$ with A over-reaching, from the perspective of the optimisation function. However, if I instead consider a bump SER function placement of 0.4, and two constituencies C and D with SERs of 4.6, then the the situations with $C = 5$, $D = 4.2$ with C overreaching, and $C = 4.2$, $D = 5$ with D overreaching are individually uniquely out of equilibrium because the proportional representation optimisation function deems the 4.2 to be unacceptable, and this kind of uniqueness is preferable. The same kind of symmetries emerge when 3 constituencies X, Y, Z with SERs of 4.33 and a bump function at $1/3$, meet at an intersection of 3 counties. While this is quite an academic point, this suggests that from an optimisation point of view, a bump position away from $1/3 = 0.333$ and $1/2 = 0.5$ is preferable. This of course must be balanced with a legal perspective on the matter, which we do not comment on here.

2.8 Multi-Objective Optimisation

In addition to the to the difficulty of properly defining a measure for each individual constraint and guideline (particularly county borders, geographic considerations, temporal continuity), there are additional problems directly to do with the nature of multi-objective objective optimisation. There is first the legal ambiguity to do with ranking the importance of each of these constraints relative to one another, and secondly there is the problem of faithfully translating this hierarchy into a numerical algorithm.

The function descriptions corresponding to each of the themes in the preceding sections are added together to give one overall optimisation function that dictates the search. However, the individual functions are not necessarily given the same weight in the overall optimisation function. We refer to the individual weights assigned to the optimisation of each legal theme the “coupling constants”. However, the technical obstacle is that there is no *a priori* way of calculating exact values for these to represent a desired hierarchy of constraints. To put it another way, it is not as though if the coupling constant on the county boundary term is increased +10% relative to the community boundary term that the search will then reflect this in a linear manner with +10% more respect for the legal theme. Determining these constants thus reduces to investigating behaviour and results of algorithms for a variety of values for each coupling constant.

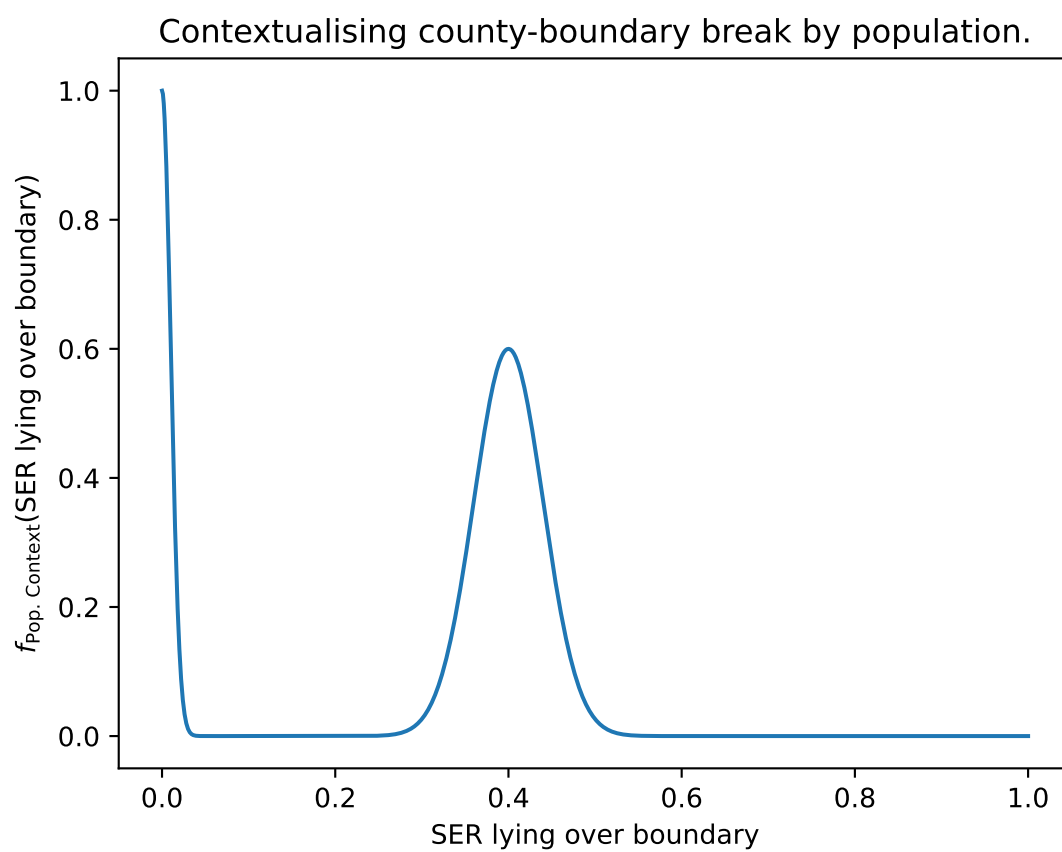


Figure 5: Caption

3 Challenges (Matthew)

The current problem is solved from a legal point of view. We will now setup the problem in such away that enables this to be treated as a math problem. But also such that it is accessible. Such that each step in the framework/process is can easily be understood.

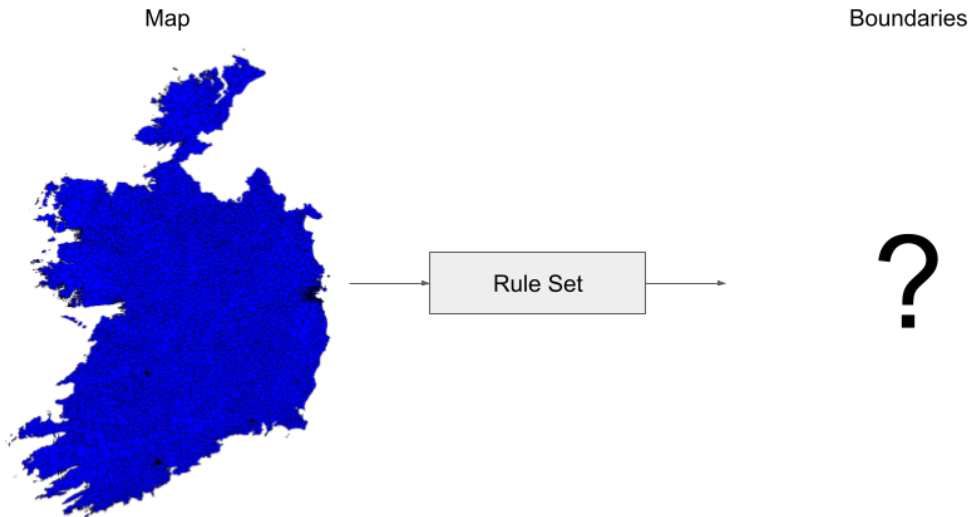


Figure 6: This diagram provides a concise visual representation of the redistricting problem as explored in this paper. It illustrates the current ambiguity involved in transforming a geographical or administrative map into valid constituencies, while adhering to a predefined "Redistricting Rule Set."

The problem of generating valid boundaries mainly boils down to one key problem. That of the combinatorial explosion of possible solutions. Attempting to traverse the space of valid solutions without the assistance of computational tools means.... (Do we want to mention they stated things they cannot verify such as the current suggestion provides minimal breaches in county boundaries). Add citation [the page in their report]

There is also the problem of interpretation/treatment of the constraints/rules.

We would argue that in an idea situation everyone would be able to reason to a similar conclusion instead of having to wait for some big "announcement day". As if a special convening of a group is required. Then the system isn't really that accessible is it... if you need the high prestes to read the n trails...

3.1 The Maths Problem

The first concept we will introduce is the term "Map". This is a collection of building blocks that make up the region which we will be redistricting. We will refer to these building blocks as districts. These districts, are the smallest components that can be used to build constituencies. Constituencies, in turn, are combinations of districts, classified as either valid or invalid based on adherence to the redistricting rule set, which defines the criteria for validity.

The challenge lies in the arrangement of districts to create valid constituencies as dictated by the rule set. This task is complicated by the intricate interplay and ripple effects that arise when relocating individual districts within the map. In the Irish case our map is the republic and the building blocks are Electoral districts (EDs). The goal is to find configurations of the EDs such that valid constituencies can be formed. The final key concept is that of "Boundaries," which comprises a collection of constituencies that meet the rule set criteria for a specific map. Notably, multiple valid boundaries can exist for a given rule set, considering the variability of seat ranges in the Irish context and or degree of acceptance for the value for the variance to name but a few. The problem of selecting which set of boundaries to go with is covered in [3.6](#)

So in short the problem is how does one arrange the districts (EDs) in such away as to generate valid constituencies. Then the question is how does one sort through the final collection of valid constituencies otherwise know as "Boundaries"? There are many steps within the overall process from Map -> New boundary, which we will outline. The methods used within the process are covered in [4](#).

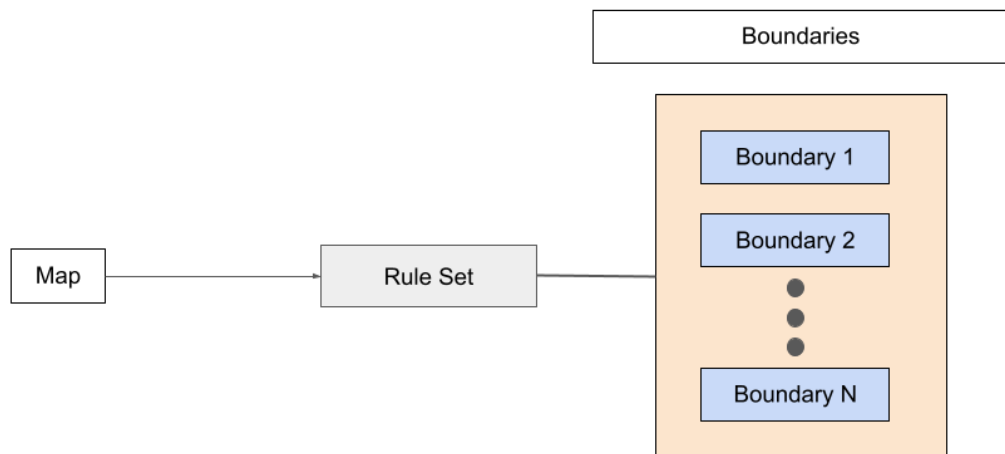


Figure 7: This diagram breaks down the process of redistricting into its core components. That of the Map, which represents the geographic or administrative region subject to redistricting. The rule set is how the aspirations of a system are codified and the result of this yields a collection of 'Boundaries', which are a collection of valid constituencies that meet criteria outlined in the rule set, allowing for multiple boundary configurations within a given rule set framework.

3.2 Outlining The abstraction

Building on the previous section, we will now outline an abstraction that will align the Irish problem with the approach we are taking. We will start with a systematic deconstruction, followed by a methodical reconstruction of the current Irish case. We find greater value in this deconstruction-reconstruction approach, as it enables us to discern the rationale behind each step, fostering a deeper understanding of the process and thus the benefits of what we are proposing.

We will start by looking at what people see from the outside looking in. That is outputs of the current redistricting process. If we examine Fig. 8 we can see the result of the current redistricting process. This is currently what people see. The boundaries being moved around to arrive at a new set of boundaries at each review (boundary review) step. The process that drives these outputs is outlined in reports associated with each review. The reports provide the recommendations and the approach taken to the rules (they are obligated to follow and those they are not). We will discuss this in detail later. But there is still yet to be an auditable trail that outlines the process either the steps or the choices that were made. Given the nature of the problem trade offs occur and some options are determined to be preferable to others (with the reasons for not being clear to outsiders).

(they mention they carried out analysis in 2023. The question might be were is this analysis to view as they make claims around how the solution they chose helps provide what I could only interpret as things one might classify as guarantees).

Constituency Commission Reports in 2017 (2, p. 16)

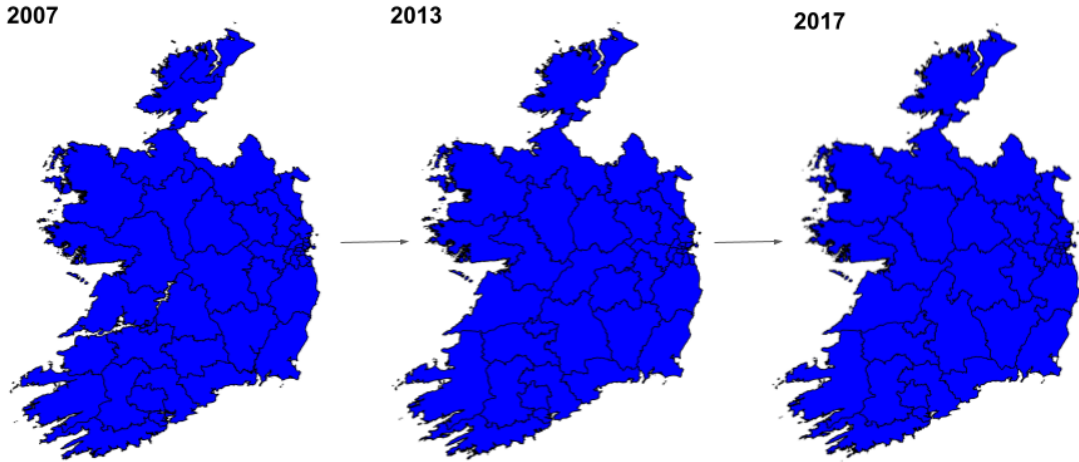


Figure 8: Evolution of Dáil Constituency Boundaries in Ireland (2007-2017): The three maps depict the recommended and accepted constituency boundaries as proposed by the Boundary Commissions over the years. The first map represents the boundaries in 2007, followed by the changes made in 2013, and the final map illustrates the updated boundaries in 2017. Arrows connect each map, showcasing the evolution of boundaries overtime.

If we break down this process to consider a singular transition within Fig. 8. In this case the transition from 2013 $\rightarrow$ 2017 and consider the generation of a single boundary (in this case 2017) Fig. 9. We still have very little information of how we get from the Map to the eventual selected boundary. But this process is happening between each iteration. With the main driving force causing the shifting of boundaries being the change in population.

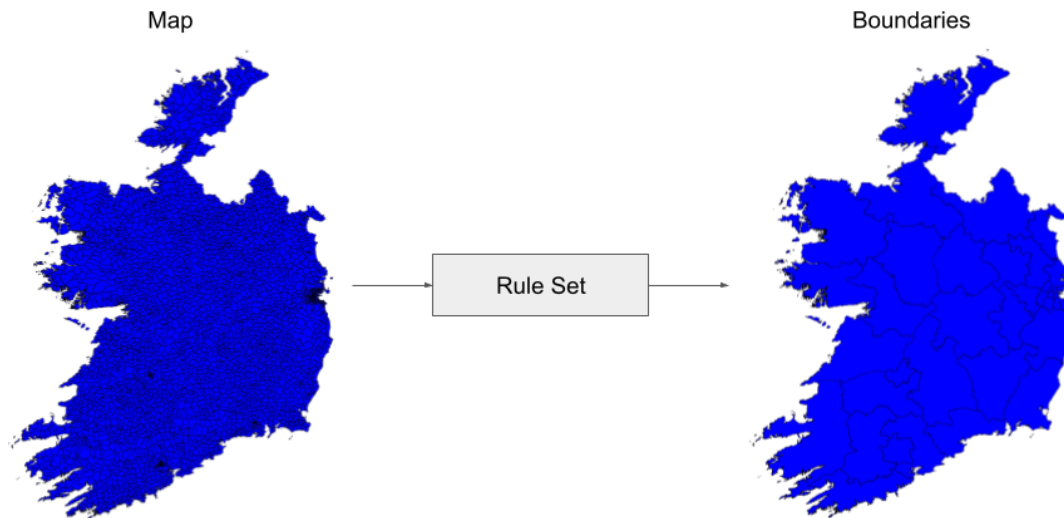


Figure 9: We see a single transition from a Map to a Boundary. In this case the recommended and accepted boundaries for 2017.

There is an issue of continuity (as technically the map doesn't start completely from scratch.) but this is discussed in another section.

Note:

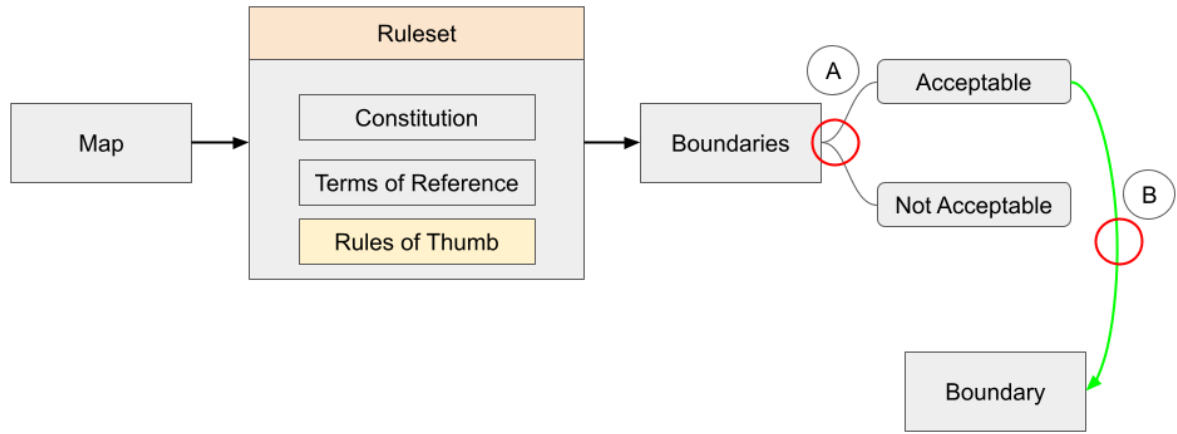


Figure 10: This figure illustrates We interpret point **A** as being the minimum population that is required to be transferred and **B** is

Diagram of Current Process

We will now move onto a diagram of the full abstraction. This is what is driving the generation of the boundaries in Fig. 8 and 9. We cannot speak to the exact process but we have broken and structured the decisions that have to be made. Each piece of this diagram represents a component in the flow. There is potential to have many different interpretations for each component. An example of this is how a particular part of the ruleset might be treated is how the electoral commission took a different approaches as stated in Constituency Report in 2023 (2, p. 19) and clearly seen in the rest of the report. One such way is the number of times they went outside the 5% with their recommendations, with over $1/3$ ($\frac{15}{43}$) of their recommendations falling outside this range. The highest percentage since the 1995 report. Each component and ones approach to them will have a significant impact on the eventual Boundary that is select (as it will impact the set of acceptable boundaries that will be considered).

- Map: We start with the EDs having their data updated. In this case it is population data. This will impact the metrics used to determine what constituencies and thus boundaries could be formed.
 - If this was part of the review process then the metrics would be calculated for the most current configuration. If needed then the process of generating new boundaries would commence.
- The rule set is now applied to the Map and the attached population data.
 - The explicitly written down rules are applied
 - The rules (more like conventions) of thumb (an example being that if two whole counties are joined it isn't considered a breach of county boundaries. At least for reports proceeding the 2023 report).

- **Boundaries:** Sets of valid constituencies under the rule set are generated to found boundaries.
 - **A:** The process of sorting through the solution space of boundaries. Boundaries are either acceptable or not acceptable. This is a part of the process that is dependent on how one interprets the ruleset. As one needs a clear and consistent and reproducible way of sorting through the boundaries. We will highlight this with examples later. One's interpretation (and in the algo case the selection of coupling constants will impact the boundaries that are valid). Your interpretation is encoded into the boundaries you find acceptable
 - **B:** The process of sorting through the set of valid boundaries and eventually selecting the boundary to be recommended/implemented.

We see that we go from some map (EDs) to the ruleset. - Each piece of this diagram represents a component in the flow. There is potential to have many different interpretations for each component

- The map is updated. New census (population data) is feed in. This then changes the metrics used to measure if a constituencies are valid (reference appendix). - This updated map is not put through the process of the ruleset being applied to it. Which involves moving around components of the Map ("EDs")

3.3 Considerations/Constraints: The Rule Set

As highlighted by Moon Duchin in her presentation in (3), understanding the universe of possibilities that is structured by the rules is essential to tackling the sort of questions we are interested in. So in this section we will be highlighting the rules/constraints that apply to the Irish case. We will highlight how they have been interpreted historically and how (if any) has their interpretation changed with time.

One thing that should be highlighted and that will be touched on throughout is realising that you are designing a system and how you think a system will behave vs what your expectations are always in agreement. Certain rules and choices will achieve certain outcomes and prevent others. This is something our framework aims to attempt to assist with. By providing people with the ability to understand the consequences of certain rules (in this case the terms of reference and constitutional requirements). A prime opportunity/example of this can be seen in the 2023 Constituency review report (2, p. 30). Where we see the commission highlight that there were issues that arose during the review. That being that they were limited to 3, 4 and 5 seat constituencies. From our point of view this is around changing the "ruleset". To do research on the issue of Constituency magnitude. They highlighted the following points

- The limitations of the current rule set: Some submissions suggest 6-seat constituencies, which are prohibited by the Act of 2022 (Current rule set).
- Others recommend revising this prohibition to allow 6 or more seat constituencies.
- Stated the potential impact of population growth on current 5-seat county constituencies in future reviews and somewhat suggesting that having the ability to have 6-seat constituencies might better enable them to meet the parts of the rule set (maintaining county boundaries)

We believe that having a framework with the appropriate machinery in place would best enable the commission to understand the impact of changing the rule set and seeing if it would actually lead to an outcome that improves what our system is

In this section we will outline in details the constraints and will touch on the theme of the need for clarification on some of the constraints and the main problem of the lack of any sort of hierarchy between them. This is in fact something that our framework and suggested algorithms could help provide a powerful tool for testing what things might look like if there was a hierarchy or different bounds for things such as variance.

While the process of translating legal requirements into mathematical definitions that can then be processed by algorithms can be a powerful tool for addressing complex issues, it is important to acknowledge that certain legal requirements may present a distinct challenge. This challenge stems from the inherent variability and subjectivity in the interpretation of these legal requirements. For instance, the interpretation of 'reasonable' or 'fair' in a legal context can vary considerably between individuals or even legal experts. Such ambiguities can introduce uncertainty when operationalising these requirements within mathematical frameworks, and it is crucial to be aware of these challenges as they can impact the effectiveness of algorithmic solutions. One can see this across the main reports by previous Electoral commissions (4), (5), (6), and (7), and now the

The constraints we are concerning ourselves with come in two forms (but are the same, one is just less flexible and stricter). Constitutional Provisions (/requirements) and Legislative (the terms of reference). They are legal definitions and have yet to be codified into operational mathematical definitions.

Constitutional Provisions and Terms of Reference in Relation to Dáil Constituencies (as highlighted in the 2023 report).

The format will be as followed

- The constraint - The metric - The history of its implementation - Both pre-electoral commission - Post electoral commission - Highlight how it's codified such that it can be handled by one of the methods (basically just reference that section)

3.3.1 Variance (representation)

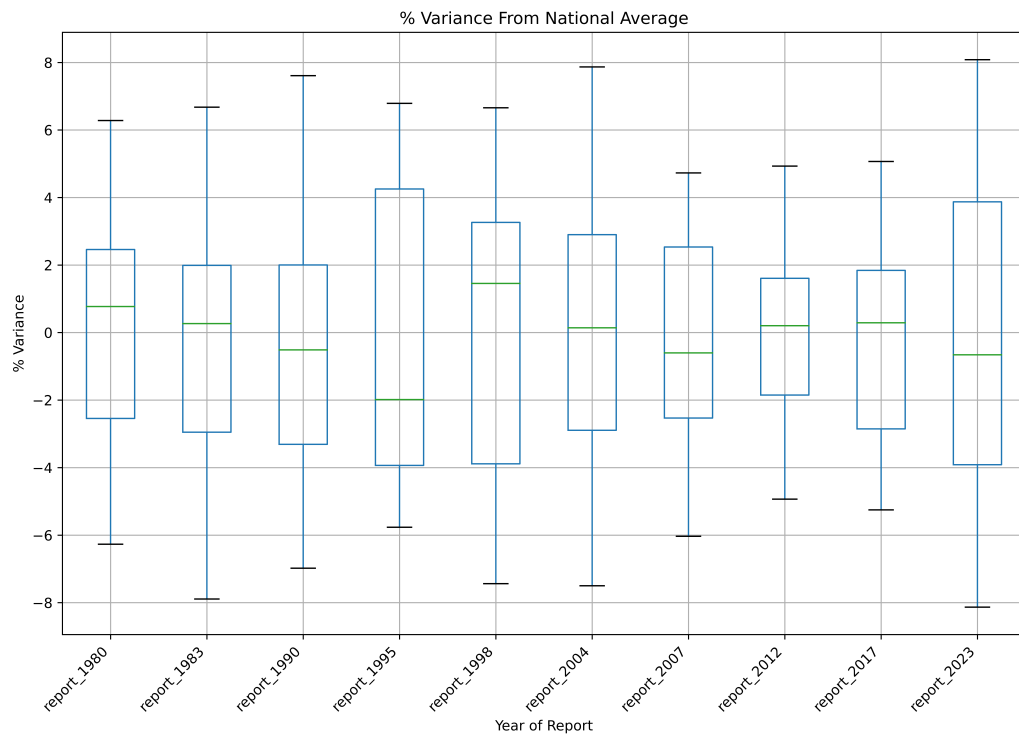


Figure 11: Box plot of the variance in representation across Constituencies from 1980-2023. The top and bottom of the error bars indicate Constituencies with the most extreme departures from equal representation for that year. The blue boxes are an indication of the spread of representation disparity (variance in the usual statistical definition).

It doesn't matter which way you cut it or from which period you look at 2023 was unprecedented when it came to the number of recommended constituencies that were outside the range.

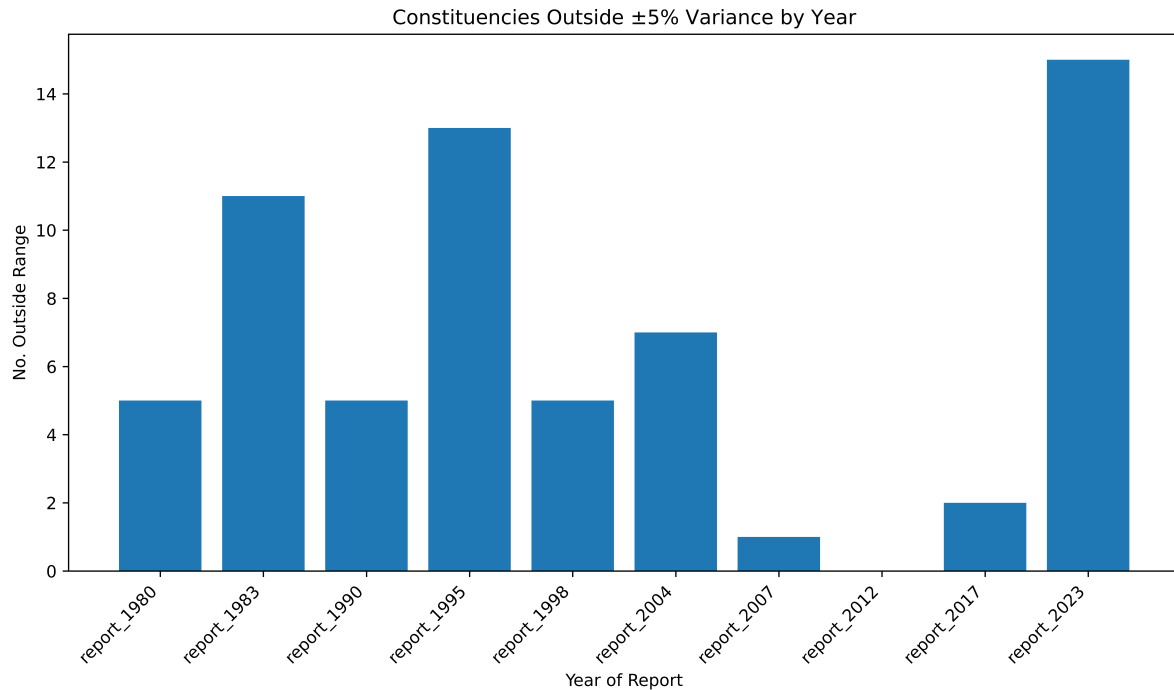


Figure 12: Bar plot showing the count of constituencies recommended outside the $\pm 5\%$ variance for each report year. The heights of the bars represent the number of constituencies with recommendations exceeding this variance threshold.

3.3.2 Continuity

When we look at how the previous reports have handled the topic of endeavouring to maintain continuity in relation to the arrangement of constituencies. We see that there has been some slight differences in the wording across the different sections from different years (7, p. 15), (6, p. 13), (5, p. 13) and (4, p. 12). But we have also seen that the overall message and themes have stayed consistent. Each report acknowledges the challenge of maintaining continuity in the arrangement of constituencies in the face of population growth and distribution, and the need to balance this with other requirements such as constitutional and geographical considerations. The idea that changes in constituency arrangements may be necessary due to population changes and the possibility of breaching county and provincial boundaries is mentioned consistently across the different sections. So, while the language and details around the discussion of continuity differ slightly between the different reports, the overall approach and view the commissions takes to it remain similar.

THE NEW WICKLOW WEXFORD CONSITUENECY. WHAT ARE THE LONG TERM IMPACTS??? AS HOW IMPORTANT WILL CONINUIITY BE WITH THIS CONSITUNECY GIVEN THE PROJECTED GROWTH OF WEXFORD AND WICKLOW. HAVE THEY DUG THEM SELFS INTO A WHOLE???

ALSO WHAT IS THE IMPACT OF EVEN MAINTING CONTINUITY ??? IF TURNOUT IS NOT A CONSIDERATION FOR DRAWING THE LINES WHY IS IT SOMETHING THAT IS CONSIDERED?.

Report Name	Number of Constituencies	Number of Constituencies Outside of Range	percentage of total recommendations outside of range
report_1980	41	5	12.20
report_1983	41	11	26.83
report_1990	41	5	12.20
report_1995	41	13	31.71
report_1998	42	5	11.90
report_2004	43	7	16.28
report_2007	43	1	2.33
report_2012	40	0	0.0
report_2017	39	2	5.13
report_2023	43	15	34.88

Table 1: Number of Historical Constituency Recommendations Outside the $\pm 5\%$ Variance Range: This table shows the number of constituency recommendations from previous Electoral Commission reports (1980 - 2023) that fell outside the $\pm 5\%$ variation from the National Average.

3.3.3 Number of Seats

- We do not see this as a constraint in the same way as the others. But more as an option that provides us flexibility. This would be the main factor we would be pushing as it is not really something that we can directly tie to something being violated (its connected to representation but that is impacted by variance. [head is to fried for this will explain to mm]

As stated in ?? the potential size of the next Dáil is set at 171-181. All our calculations take into consideration this full range. But as will become clear as one examines the numbers, the range of 178 - 181 seats is the only range that allows for boundaries that make sense, when accounting for the terms of reference. Within all our recommendations, we provide a range for which our recommendation is valid.

3.4 Searching The Solution Space

The problem from a technical point of view is searching the space of possible configurations and then finding valid solutions. The number of possible ways in which the building blocks (EDs) can be arranged to form constituencies is beyond substantial. Even if we search a small area, its more than a human could traverse.

"even when winnowing down to districting plans that satisfy criteria set forth by legislatures, commissions, or voter referenda, the number remains far too large to enumerate all possible plans in a state. The numbers are easily in the range of googols rather than billions, as we will explain here."

- insert diagram "Recent methods for analyzing and comparing districting plans attempt to do so by placing a plan in the context of valid alternatives—that is, those that cut up the same jurisdiction by the same rules and with the structural features of the geography and the pattern of voting held constant. Modern computational techniques can generate large and diverse ensembles of comparison plans, even if building the full space of alternatives is out of reach. These ensembles contain samples from the full space of plans, aiming to help compare a plan's properties to the range of possible designs. More than this, we need some assurance of representative sampling; that is, we need to relate the sampling distribution to the rules and priorities articulated by redistricters." "But the inferences that can be drawn from ensembles rely heavily on the distributions from which the ensembles are sampled."

We have opted for two options:

1. MCMC for sampling. "methods offer strong underlying theory and heuristics, in the form of mixing theorems and convergence diagnostics." "The idea is to form a random walk on the state space and collect states visited by the walker to build a sample. The first and most natural approach to defining a random walk on graph partitions is to simply reassign one node at a time through a random process in a process. In a districting context, this amounts to changing the labeling of individual geographic units along district borders. In the standard MCMC paradigm, we would modify the basic step to adjust the ultimate stationary distribution. The traditional ways of doing that are explored here."
2. Cas's approach

3.5 Key Issues

One of the issues that arises with the ruleset in the Irish case is the lack of clarity on the

As we currently see it, a good system of rules should enable people to work to a solution that is very close to the final accepted one without too much effort. But we would argue this is not the case with the current rule set.

3.6 A Good Map/Boundary

- There will inevitably be multiple options and the question is which one ends up being the one selected and what is the basis for this

3.7 Proportional Representation at the Ballot vs. Accurate Representation during term

As stated in the outset, PR aims to reflect the political distribution of the electorate in the makeup of the elected representatives. However, when paired with the practical need for constituencies, this perhaps introduces a fundamental tension which cannot be resolved. It has

been acknowledged that the larger the number of seats in a constituency, the greater the opportunity for representatives of smaller parties to make it in. This points to larger constituencies facilitating a greater resolution into the true, underlying distribution of political opinions in the electorate, and that smaller seat constituencies coarsens the reflection of said distribution in the representatives, favouring larger parties. However, as is demonstrable using the most recent election data, larger constituencies increasingly enable disproportionate geographical representation. (Matthew's plot) Representatives can cluster geographically in one section of their constituency. This poses an interesting, open question: If PR is supposed to facilitate a faithful, proportionate embodiment of the constituents' political beliefs in their representatives, do we content ourselves with greater fidelity at the ballot at the expense of exaggerated uneven access to representatives during the term?

4 Methodology

4.1 Models

4.1.1 Potts Model

The Standard Potts Model

In his 1951 PhD thesis, Australian mathematician Renfrey Potts introduced a model (8) of interacting spins, now aptly known as the Potts model. The model consists of a lattice of spin sites s_i , each of which are assigned to one of Q values. The interactions of the spins are determined by a function known as the Hamiltonian H , in that a different Hamiltonian will result in different dynamics of the model. The standard Potts model is defined by the Hamiltonian

$$H_{\text{Potts}} = -J_{\text{Potts}} \sum_{\langle i,j \rangle} \delta_{s_i, s_j}, \quad (9)$$

where the coupling constant J_{Potts} determines the strength of interactions, the summation is over all nearest-neighbour pairs $\langle i, j \rangle$, and the Kronecker delta $\delta_{a,b}$ is defined as

$$\delta_{a,b} = \begin{cases} 1, & a = b \\ 0, & a \neq b \end{cases}. \quad (10)$$

In the standard Potts model, spins only interact with a nearest neighbour based on whether or not their values are equal, where pairs of equal sites reduce the Hamiltonian and pairs of unequal sites do not contribute. In this sense, the values taken by each spin need not be numerical, but can be elements of any abstract set of size Q ; for example, $\{1, 2, 3\}$ and $\{\text{red}, \text{green}, \text{blue}\}$ are equivalent sets of values for the $Q = 3$ standard Potts model.

It is straightforward to verify that H_{Potts} is minimised when all lattice sites are assigned to the same value, i.e. when $s_i = s_j$ for all i, j ; this is known as a ferromagnetic configuration. If the Hamiltonian was defined differently, then a different configuration would minimise it; for example, since an antiferromagnetic configuration ($s_i \neq s_j$ for all neighbours i, j) maximises H_{Potts} , then it must also minimise the Hamiltonian $H = -H_{\text{Potts}}$.

The Electoral Potts Model

The first step in characterising the Electoral Potts model (9) is to describe the redistricting problem using the language of the Potts model; the 3,440 Irish EDs assigned to Q constituencies¹ are treated as 3,440 lattice sites assigned to Q spin values, constituency boundaries are determined by the groupings of these values, and geographical ED borders translate directly into nearest-neighbour lattice sites.

The next step is to encode the desired legal requirements into separate Hamiltonian functions, such that adhering to a requirement minimises the corresponding Hamiltonian. The main requirements for a configuration of constituencies are Proportional representation (subsection 2.2), Contiguity (subsection 2.3), and compactness² (Dlúthacht as Gaeilge) (subsection 2.4), represented by the Hamiltonians

$$H_P = \sum_{q=1}^Q \min_{m \in \{m_s\}} \left| \frac{P_q}{m} - \langle P \rangle \right|, \quad H_C = \sum_{q=1}^Q |G_q - 1|, \quad H_D = \sum_{\langle i,j \rangle} (1 - \delta_{q_i, q_j}), \quad (11)$$

¹The 2023 Constituency Review Report recommends an increase from $Q = 39$ to $Q = 43$.

²In this model, compactness is defined based on the length of a constituency's border with another constituency.

where P_q and $\langle P \rangle$ are the population of constituency q and average population per constituency respectively, $\{m_s\}$ is the allowed numbers of seats per constituency,³ G_q is the number of contiguous groups in constituency q , q_i is the constituency to which district i is allocated, and $\langle i, j \rangle$ denotes neighbouring districts i and j . It is straightforward to verify that

- H_P is minimised when each constituency represents an allowed integer multiple of the average population per constituency ($P_q = m\langle P \rangle$ for some $m \in \{m_s\}$, for each $q = 1, \dots, Q$; this is illustrated in Figure 13),
- H_C is minimised when each constituency is made up of exactly one connected group of EDs ($G_q = 1$ for each $q = 1, \dots, Q$; note that the modulus $||$ is used so that a constituency q contributes to H_C even when no ED is assigned to it, i.e. when $G_q = 0$), and
- H_D is minimised when all EDs are assigned to the same constituency ($q_i = q_j$ for all i, j ; note that $H_D = \frac{H_{\text{Potts}}}{J_{\text{Potts}}} + \sum_{\langle i, j \rangle} 1$, and so is also minimised by a ferromagnetic configuration).

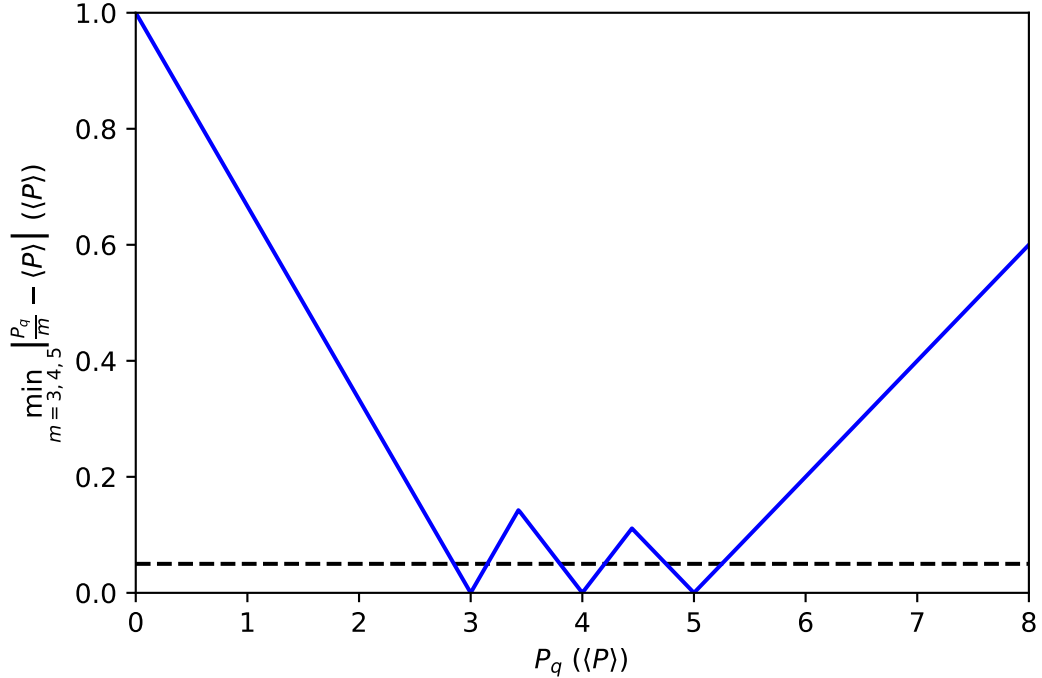


Figure 13: Contribution of a single constituency to H_P against constituency population, for $\{m_s\} = \{3, 4, 5\}$. Black dashed line represents a 5% variance.

Implementing other considerations would amount to defining corresponding Hamiltonian terms in a similar manner. However, vaguely defined constraints might require further clarification;

³The Electoral Act of 1997 specifies $\{m_s\} = \{3, 4, 5\}$, i.e. 3-, 4-, and 5-seater constituencies.

for example, each of

$$\begin{aligned}
H^{(1)} &= \sum_{q=1}^Q (\mathcal{C}_q - 1), \\
H^{(2)} &= \sum_{q=1}^Q (1 - \delta_{\mathcal{C}_q, 1}) \sum_{c_q} (1 - \delta_{\pi_{c_q}, 1}), \\
H^{(3)} &= \sum_{q=1}^Q \sum_{\pi_{c_q} \neq \pi_{c_q, \max}} \frac{1 - |2\pi_{c_q} - 1|}{2}
\end{aligned}$$

(where $\mathcal{C}_q$ is the number of counties c_q with EDs in constituency q , and π_{c_q} is the proportion⁴ of county c_q contained within constituency q) could be seen as adequate Hamiltonians for county borders, depending on the interpretation of a “county border violation” (subsection 2.5).

The final step is to form a linear combination of these Hamiltonians to create an objective function

$$H = \sum_{\alpha} J_{\alpha} H_{\alpha}, \quad (12)$$

where $\alpha = P, C, D, \dots$ refers to a requirement/guideline and the coupling constant J_{α} implicitly determines the consideration given to that requirement/guideline. Through this modified model, the redistricting problem has now been reduced to

- translating the desired hierarchy of rules and guidelines $\{\alpha\}$ into numerical coupling constants $\{J_{\alpha}\}$, and
- determining the configuration of constituencies that minimises Equation 12 for a given set of constants $\{J_{\alpha}\}$.

The first of these objectives requires exploration of the coupling constant parameter space, i.e. comparing results for variety of combinations of $\{J_{\alpha}\}$, and is further discussed in subsubsection 4.2.2. The second objective is simply an optimisation problem which, similarly to how the Electoral Potts model is based on a model in condensed matter physics, can be tackled using methods popular in statistical mechanics.

Configuration Optimisation

From inspection of the constraint Hamiltonians (Equation 11) it is easy to determine a configuration of constituencies that minimises H_C (each constituency is contiguous) or H_D (ferromagnetic), however finding one that minimises H_P is a non-trivial problem. Furthermore, minimising the objective Hamiltonian H (Equation 12) for a given set of coupling constants $\{J_{\alpha}\}$ is even more difficult. It is computationally infeasible to calculate H for each unique configuration of constituencies,⁵ and so numerical approximation methods are required to arrive at sufficiently optimal electoral maps.

In statistical mechanics, the (discrete) Boltzmann distribution⁶

$$\lambda(\sigma) \propto \exp\left(-\frac{H(\sigma)}{T}\right) \quad (13)$$

⁴Furthermore, the definition of proportion could be defined based on population, area, or number of EDs.

⁵ $S(3440, Q) \gg 10^{5000}$ for $Q \geq 30$.

⁶For simplicity, $k_B = 1$ (or equivalently, $k_B T \rightarrow T$).

represents the probability of a system at temperature T being in a state σ with energy $H(\sigma) \equiv H$. By comparison to the (continuous) exponential distribution

$$f_{\text{Exp}}(x) \propto e^{-\beta x} \quad (14)$$

as depicted in [Figure 14](#), it is clear that, as $T \rightarrow 0$, $\lambda(\sigma)$ approaches a uniform distribution of the states with minimum energy. As such, if this distribution can be adequately sampled from, then doing so for a small temperature T will return configurations that minimise H .

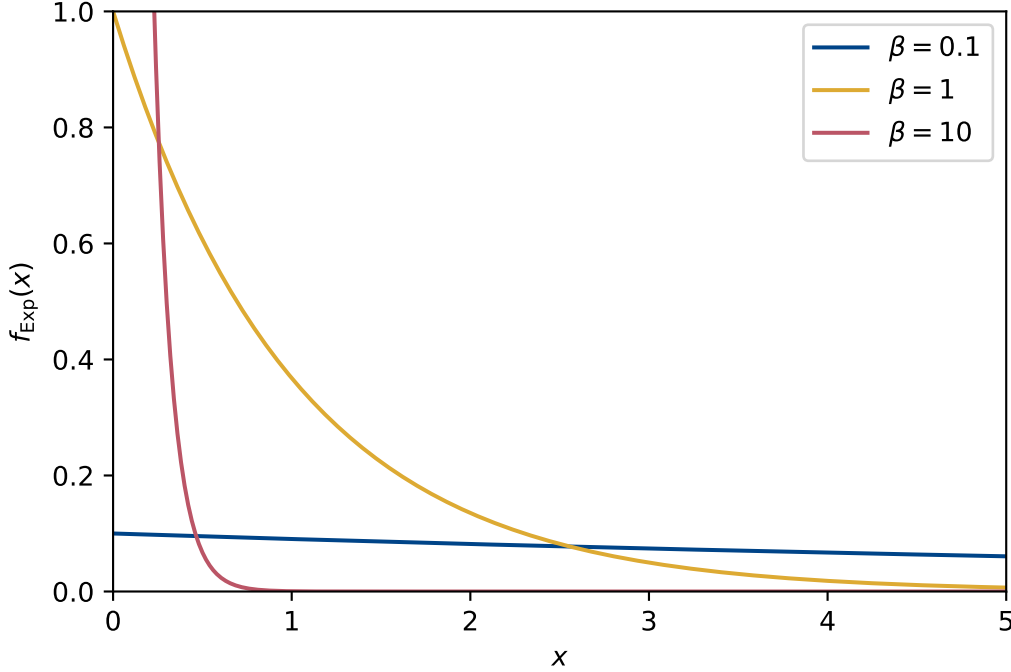


Figure 14: Exponential distribution ([Equation 14](#)) for $\beta = 1, 5, 25$.

While mainly used to numerically approximate multi-dimensional integrals, Markov chain Monte Carlo (MCMC) methods are algorithms that sample from a desired probability distribution. They rely on random sampling (Monte Carlo) from a simpler distribution depending on the previous sample to form a sequence of samples (Markov chain) from the desired distribution. Two examples of MCMC methods are the Metropolis algorithm and the Gibbs sampler, and can be used to sample configurations from $\lambda(\sigma)$ for a fixed temperature T as follows:⁷

⁷See [Appendix D](#) for an introduction to MCMC methods and a motivation for the Metropolis algorithm and Gibbs sampler.

Metropolis algorithm

1. Choose an ED i and record its constituency q_i .
2. Choose a different constituency $q'_i \neq q_i$ at random.
3. Calculate the change ΔH in the Hamiltonian H due to a proposed update $q_i \rightarrow q'_i$.
4. (a) If $\Delta H \leq 0$, update $q_i \rightarrow q'_i$.
(b) If $\Delta H > 0$, update $q_i \rightarrow q'_i$ with probability $\exp(-\frac{\Delta H}{T})$.
5. Repeat steps 1-4 for each ED.

Gibbs sampler

1. Choose an ED i and record its constituency q_i .
2. Calculate the changes $\Delta H^{(q)}$ in the Hamiltonian H due to updating $q_i \rightarrow q$ for each $q = 1, \dots, Q$ (note that $\Delta H^{(q_i)} = 0$).
3. Choose q'_i according to the distribution
$$\frac{\exp\left(-\frac{\Delta H^{(q'_i)}}{T}\right)}{\sum_{q=1}^Q \exp\left(-\frac{\Delta H^{(q)}}{T}\right)}.$$
4. Update $q_i \rightarrow q'_i$.
5. Repeat steps 1-4 for each ED.

By initialising a random constituency configuration and employing a technique called simulated annealing ([subsubsection 4.2.1](#)), these methods can be used to sample from $\lambda(\sigma)$ at small temperatures T , thereby obtaining optimal electoral map configurations for a given number of constituencies Q and coupling constants $\{J_\alpha\}$.

4.1.2 Genetic Algorithm

One need only look at the world around us to appreciate the fact that evolution by natural selection has resulted in some of the most remarkably complicated structures operating at near optimal efficiencies in their environment. Oversimplifying the process, we can describe evolution as a 3 step process that turns what is effectively randomly generated beings - unfit for their environment, into their most optimal form.

1. A large group of **individuals** (called the **population**) fights amongst themselves in their environment such that the population is "ranked" in order of "strength" or more accurately, **fitness**.
2. This group of top individuals are picked out and given the highest chance of successfully breeding.
3. Breeding causes a new **generation** to come about. This acts as the new population and Step 1. is repeated.

This basis provides a good starting point to designing a computational algorithm to best use the principles of evolution to generate optimal solutions to a problem.

In this section, we analyse the custom designed Simulated Annealing - Genetic Algorithm (**SAGA**) method. Combining the aforementioned layout of a genetic algorithm with the control regime set out by simulating the metallurgical process of annealing as discussed in [subsubsection 4.2.1](#).

The first step is the creation of an initial population, built up of a set of individuals. First and foremost, the question is "how can we phrase the problem of drawing constituencies (or seat allocations) such that we can create a set of individuals with a quantifiable concept of fitness?". The answer to which is to identify each individual with a **configuration** of the given EDs i.e. every ED is given one of Q values, where Q is the number of constituencies or seats. This is a working representation of the problem in terms of individuals which can be grouped together to form a population. We now focus on the form of the fitness calculation. We first identify

the aspects of a configuration which we would like to optimise: contiguity / compactness and population variance are the most fundamental. So we say that the total fitness of a configuration is said to be a linear combination of how "good" the configuration's compactness and adherence to population variance is i.e.

$$F = \gamma_{\text{comp}}D + \gamma_{\text{pop}}P \quad (15)$$

Looking first at compactness, we use an area of mathematics called Graph Theory. We generate a graph (add in example graph) wherein each ED is a node. We say that node A, which corresponds to ED A, is connected to node B iff nodes A and B share a geographical border and belong to the same seat allocation. We then count the amount of edges, or connections, in this graph and call this E . The total amount of edges possible in a graph with N nodes, is the edges of the complete graph of N nodes which is $N(N - 1)/2$ [reference?]. This is obviously impossible as it would require every ED to geographically border every other ED. If there are N EDs in the area being analysed we say $N(N - 1)/2 = E_{\text{tot}}$. To obtain the desired behavior the following compactness function was constructed

$$D = \kappa \arcsin\left(\frac{E}{E_{\text{tot}}}\right) \quad (16)$$

Where κ is a constant of proportionality. Given that the ratio $\frac{E}{E_{\text{tot}}}$ is quite small, we have that the difference between the compactness of two similar configurations A and B, $\arcsin(\frac{E_A}{E_{\text{tot}}}) - \arcsin(\frac{E_B}{E_{\text{tot}}}) \approx \frac{E_A - E_B}{E_{\text{tot}}}$. This is quite a small number, so a constant κ was introduced such that the difference is more pronounced. For this case we set $\kappa = \frac{50}{3}$.

The other term in this fitness is that which corresponds to the level of conformity to the population variance found in the configuration. This is a relatively simple calculation originating from the behaviour of the used mathematical functions. Namely,

$$P = \left[1 - \frac{2}{\pi} \arctan\left(\frac{1}{Q} \sum_{i=1}^Q \left|\frac{\mathcal{P}_i}{\mathcal{P}_{\text{avg}}} - 1\right|\right)\right]^\eta \quad (17)$$

Although computationally simple, this is not a simple expression so to speak. Here, as before, Q is the number of constituencies or seats. There are three unseen variables here. The η term is defined to be the *harshness*. It allows us to control how sharply the punishment for configurations that do not meet a certain level of variance changes. That is to say, for low η values if a new configuration is suggested where the population variance is relatively far away from the minimal variance (zero) then this can be let slide. However, in high η regimes, even a slight movement away from minimal variance causes a large drop in the P term of the configuration's fitness. This is not to be confused with the relative importance of the population variance term to the compactness term, which is given by γ_{comp} and γ_{pop} . In this case, $\eta = 5$ was used. The other two previously unseen variables are the terms $\mathcal{P}_i$ and $\mathcal{P}_{\text{avg}}$. These correspond to the population of each proposed constituency or seat and the average population of all the given space, i.e. $\mathcal{P}_{\text{avg}} = \frac{1}{Q} \sum_{j=1}^Q \mathcal{P}_j$, respectively.

Now that we have fully developed the fitness function for our genetic algorithm, we turn our heads to actual carrying out of said algorithm. An initial temperature, T_0 , is chosen which is to decrease geometrically in each generation by a factor of 0.99. That is to say, the temperature at generation k is given by $T_k = (0.99)^k T_0$. This value will then be used to carry out the simulated annealing aspect of SAGA. In this case, an initial population of size 2 is created and each individual is treated sequentially and goes through the same process separately. The individual is firstly **mutated**, in this case that just means that an ED is picked, at random, and another ED is picked

4.2 Techniques

4.2.1 Simulated Annealing

The optimisation methods for the two models presented here each rely on an objective function to be minimised (Equation 12) or maximised (Equation 15) and an external “temperature” parameter that determines the behaviour of the corresponding algorithm. Setting this temperature parameter at a high value results in uniform random sampling of the entire space of constituency configurations (Figure 15a), while setting a low temperature restricts sampling around a local extremum near the initial configuration (Figure 15b); in either case, the probability of obtaining the desired global extremum is extremely small. If the temperature was instead initially set high and slowly reduced over time, this would ensure that the algorithm would always be sampling from the temperature-dependent equilibrium distribution, until it reached low temperatures and thus the desired extremum corresponding to the most optimal configuration (Figure 15c). This approach is called simulated annealing (10, 11), named after the technique in materials science of heating and cooling a material in a specific way to change its atomic structure and thus its physical properties.

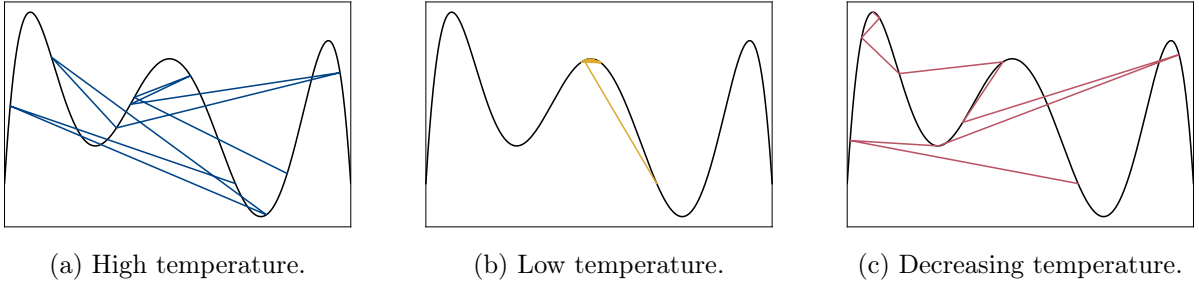


Figure 15: Examples of a temperature-dependant algorithm maximising a one-dimensional function for fixed high and low temperatures (similar to uniform random sampling and hill climbing, respectively) and decreasing temperature (simulated annealing).

4.2.2 Coupling Constants

5 Results

5.1 Small-Scale Problem

(This paper does not directly approach the full-scale problem of redistricting Dáil constituencies, but rather focuses on the similar problem of allocating seats within a constituency. This was done to a) verify the legitimacy of the outlined frameworks, b) experiment with constraint applications and multi-objective optimisation, and c) locate and adjust algorithm performance bottlenecks, all while avoiding the higher computational expense of the full-scale problem. The seat allocation problem acts as a proof of concept of the constituency allocation problem, as they each represent the same task of forming groups of EDs to follow custom constraints, albeit at different scales. Might compare Carlow-Kilkenny and Ireland via distributions of ED population and connectivity - should probably be similar shapes so can say that analysis and investigation is similar.)

5.1.1 Potts Results - Ruaidhrí

(normalised constraint Hamiltonians to be bounded between 0 and 1, for easier plotting and comparing of each Hamiltonian and also easier to translate constraint hierarchy into coupling constant values. constraint Hamiltonians (Equation 11) $H_\alpha \rightarrow \frac{1}{Z_\alpha} H_\alpha$, where

$$Z_P = 2(Q-1)\langle P \rangle, \quad Z_C = \sum_i 1 - Q, \quad Z_D = \sum_{\langle i,j \rangle} 1 \quad (18)$$

are normalising factors. note: $-Q$ in Z_C not included in code, but minimal effect as 167 instead of 162. also have $\{m_s\} = \{1\}$ for H_P as only considering single-seat groupings.)

(notice that scaling coupling constants and temperature by same factor has no effect on either algorithm, i.e. we have the symmetry $T \rightarrow \kappa T$, $J_\alpha \rightarrow \kappa J_\alpha$ for some $\kappa > 0$, as $\Delta H \leq 0 \implies \kappa \Delta H \leq 0$ and $\frac{\Delta H}{T} = \frac{\kappa \Delta H}{\kappa T}$. can (a) remove this degeneracy (ensure uniqueness of set of external parameters) and (b) reduce degrees of freedom (number of external parameters) simply by picking a coupling constant (that we know would never be 0) and only considering the ratio of relevant terms to this constant (equivalent to just setting this constant to be 1, or setting κ as the inverse of this constant). self-righteously determined that proportional representation is the most important constraint ($J_P \neq 0$), and so picked this as the guinea pig. thus the external parameters are $\frac{J_C}{J_P}$ and $\frac{J_D}{J_P}$, the change $\frac{\Delta H}{J_P}$ is calculated to accept and reject proposed configuration updates, and simulated annealing varies $\frac{T}{J_P}$ from high to low)

(initial temp for given J_C, J_D was chosen such that any proposal in the Metropolis algorithm had at least a 99% chance of being accepted, i.e.

$$\exp\left(-\frac{\Delta H_{\max}}{T_0}\right) = 0.99 \implies \frac{T_0}{J_P} = \frac{\Delta H_{P,\max} + \frac{J_C}{J_P} \Delta H_{C,\max} + \frac{J_D}{J_P} \Delta H_{D,\max}}{-\ln 0.99}. \quad (19)$$

Subsequent temperatures were found by scaling by a factor of 0.9, i.e. $T_k = 0.9^k T_0$, for a total of 150 temperatures)

5.1.2 GA results - Cas

5.2 Further Work

5.2.1 Extending to Full-Scale Problem

5.2.2 Exploring the Parameter Space

5.2.3 Other Potential Use Cases

We will highlight a few ways in which this framework can be utilised and what value it could potential provide to decision makers and the public/stakeholders.

6 Conclusion

6.1 Algorithmic Redistricting Framework

Benefits of Algorithmic Approach

To put it simple the problem is that of generating valid maps as defined by the rule sets that an area writes to decide how their districts are drawn. In this paper, we employ mathematical computational techniques to tackle the complex problem of defining valid electoral districts based on a set of rules designed to codify aspirational goals within an electoral system. We want to emphasise that our work is not intended to make assertions about the correctness or morality of any specific electoral districting solution. Instead, we aim to provide a mathematical framework that assists in the evaluation and analysis of the Irish electoral system in alignment with desired values.

One such value we consider is the principle of equal representation, among others. We do not prescribe a specific definition of representation, as it is inherently a normative concept, but rather, we provide the tools necessary for individuals and policymakers to assess whether their proposed rule sets can lead to systems that align with their values.

Our belief is grounded in the understanding that addressing normative questions effectively often requires a deep understanding of the mathematical implications and impacts of the chosen rules. By leveraging mathematical machinery, we offer a means for stakeholders to quantitatively assess the consequences of their decisions and gain insights into how different rule sets may influence outcomes.

While many nations grapple with issues related to electoral districting, Ireland faces a distinct set of challenges. Unlike the United States, where gerrymandering has been a prominent concern, the problem Ireland confronts is one of efficiency and ensuring that the principles of proportional representation are upheld.

In this paper, we aim to address the specific challenges posed by electoral districting in Ireland. Our primary objectives are twofold: to enhance the efficiency of the electoral districting process and to explore the potential of computational methods in achieving fair representation. We see this endeavor as an opportunity to introduce innovative techniques, previously applied in different jurisdictions and systems, to the Irish electoral landscape.

As we delve into the intricacies of electoral districting, we aim to harness the power of computational methods and mathematical modeling to design impartial districts, assess existing constraints, and test the resilience of the rules against gerrymandering.

This paper does not aim to provide definitive answers but serves as a guide and a tool for decision-makers. It offers insights into the mathematical underpinnings of electoral districting, explores the axioms and criteria governing the process, and raises questions about the system's limitations. As we delve into this exploration, we hope to contribute to the ongoing discourse on electoral fairness and representation in Ireland. Our work hopes to show the benefits of mathematics in shaping a more representative and equitable electoral system.

our work serves as a mathematical toolset, providing valuable assistance to those seeking to navigate the normative complexities of electoral districting. We facilitate the exploration of how various rules and parameters can impact the realization of desired electoral system characteristics, without making normative judgments about what those desired characteristics should be.

The decision support this framework provides could be critical, considering the significant challenge the commission faces in reviewing and potentially redrawing electoral boundaries at various levels, including EU, LEA, Dáil, and more. Especially if they all coincide together.

6.2 Implementing: MCMC for Redistricting

- We are focused on the problem of sampling from the STATE SPACE OF....

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A Data

This section outlines how key data for this problem of redistricting in the Irish context was collected and the general issue open data. - You should not be required to have specialist knowledge to access the data necessary to answer a question

- There are EDs and then there are CSO EDs. CSO EDs are the level at which data is reported. So need to note this.

A.1 Variance Data

To calculate the variance as defined in equation 6 as discussed in 2.2, one requires the following bits of information:

- Number of Dáil seats (members) per constituency

But for the variance calculation per proposed constituency it is easier to make use of the following columns provided by the CSO int

- Number of Dáil seats (members) per constituency
- In most cases this data was directly calculated from tables from the CSO CSO's PxStat Open Data Statistical Database.
- In other cases we needed to go directly to the Dáil Constituency Commission reports to pull the tables **Statistics Relating to the Proposed Constituencies (12)** . This table is standard across all reports with the following data being given
 - Third level item 1990
 - Third level item 2
- Also note that we have
- Columns
 - Third level item 1990
 - Third level item 2

For variance data older than 1990 you have to look the EXPLANATORY MEMORANDUM (13) and (14) where the variance column was called "Variation from national average population per member" instead of the 2023 name of "variance from National Average Population per TD". In the explanatory memorandum, it must be noted that column was reported in the number of people instead of the number of people/national average * 100

A.2 Boundary Data

B Considerations

In this section of this submission, we will outline several key points that form the foundation of our positions and arguments. These points are critical to understanding our perspective and reasoning.

C Alternative Definition for Variance

This excerpt is taken from our team’s recent boundary submission, and put here for completeness sake. There is an alternative functional for variance, the output of which varies only slightly from the previous version, but it is formulated in a more voter- and representation-central manner. Let “variance” be denoted by v , “seat equivalent representation” be denoted by s , and “assigned seats” be denoted by a . We have:

$$v_{\text{old}}(s) = \frac{s - a}{a} \quad v_{\text{new}}(s) = \frac{s - a}{s}$$

What is the effect of swapping the variable in the denominator? To illustrate this, let us consider the maximum and minimum s afforded by the old definition of variance, for fixed a , and v .

$$\begin{aligned} v &= \frac{s_{\text{max}} - a}{a} & -v &= \frac{s_{\text{min}} - a}{a} \\ \implies s_{\text{max}} &= a + va & \implies s_{\text{min}} &= a - va \end{aligned} \tag{20}$$

The problem with the old definition is most obvious when we speak in terms of population, so we rewrite the equations by multiplying by the National Ratio, as in Eq. 5.

$$\implies p_{\text{max}} = p + vp \quad \implies p_{\text{min}} = p - vp \tag{21}$$

Now for a maximally under-represented population, we find the number of TDs per constituent, and find the discrepancy with the number of TDs per constituent in the ideal, national average case. We do the same for a constituent in a maximally over-represented population. In the old definition of variance, we find that the overall mis-representation (absolute value of representational discrepancy) of extremal populations in the under- and over-represented cases **are not equal**. In other words, the bounds provided by the old definition of variance do not signify the same degree of misrepresentation.

$$\begin{aligned}
\text{Misrep.}(p_{\max}) &= \left| \frac{a}{p_{\max}} - \frac{a}{p} \right| & \text{Misrep.}(p_{\min}) &= \left| \frac{a}{p_{\min}} - \frac{a}{p} \right| \\
&= \left| \frac{a}{p+vp} - \frac{a}{p} \right| & &= \left| \frac{a}{p-vp} - \frac{a}{p} \right| \\
&= a \left| \frac{p-(p+vp)}{(p+vp)p} \right| & &= a \left| \frac{p-(p-vp)}{(p-vp)p} \right| \quad (22) \\
&= a \left| \frac{-vp}{p^2(1+v)} \right| & &= a \left| \frac{vp}{p^2(1-v)} \right| \\
&= \frac{av}{p(1+v)} & &= \frac{av}{p(1-v)}
\end{aligned}$$

As we can see, they are not equal. Let us follow the same procedure with the alternative, new definition for variance; we find the extremal populations, and contrast the extent of the misrepresentation of constituents in such over- and under-represented populations.

$$\begin{aligned}
v &= \frac{s_{\max} - a}{s_{\max}} & -v &= \frac{s_{\min} - a}{s_{\min}} \\
\Rightarrow s_{\max} &= \frac{a}{1-v} & \Rightarrow s_{\min} &= \frac{a}{1+v} \\
\Rightarrow p_{\max} &= \frac{p}{1-v} & \Rightarrow p_{\min} &= \frac{p}{1+v}
\end{aligned}$$

$$\begin{aligned}
\text{Misrep.}(p_{\max}) &= \left| \frac{a}{p_{\max}} - \frac{a}{p} \right| & \text{Misrep.}(p_{\min}) &= \left| \frac{a}{p_{\min}} - \frac{a}{p} \right| \\
&= a \left| \frac{1-v}{p} - \frac{1}{p} \right| & &= a \left| \frac{1+v}{p} - \frac{1}{p} \right| \quad (23) \\
&= a \left| \frac{1-v-1}{p} \right| & &= a \left| \frac{1+v-1}{p} \right| \\
&= a \left| \frac{-v}{p} \right| & &= a \left| \frac{+v}{p} \right| \\
&= \frac{av}{p} & &= \frac{av}{p}
\end{aligned}$$

Under the new definition of variance, we find the populations at the bounds experience the same degree of misrepresentation, which is much more transparent. The generality of the derivation means this result holds for any a -seater constituency or choice of threshold, v . This happens because by putting the deserved constituency representation, (i.e. the seat equivalent

representation, s) in the denominator, deserved representation acts as the reference, rather than assigned representation.

C.1 Heuristics for Assigning Constituencies from Past Reports

Joining two entire counties

As stated in previous Constituency Commission Reports in 2017 (7, p. 16), 2012 (6, p. 13), 2007 (5, p. 12) and (4, p. 11) the Commission has adhered to the view and practice of previous commissions namely that joining two entire counties in a single constituency does not constitute a breach of the boundary between them. This is an important precedent that we will be using thought our submission. As we assume this view and practice will be continued by this commission.

Balance Between The Three Constituency Sizes

As stated in previous Constituency Commission Reports in 2017 (7, p. 15) and the report in 2012 (6, p. 13) it is not a constitutional or statutory criterion that the arrangement of constituencies should seek to achieve any particular number of, or balance between, the three constituency sizes.

Importance of Provincial Boundaries

Within previous commissions reports and their terms of reference (4),(5),(6), and(7), there has not been any explicit reference to the consideration of maintaining provincial boundaries.

Previous Constituency Commission reports have not explicitly stated the importance of maintaining provincial boundaries. However, in earlier reports such as those in 2002 and 2007, the Commission did make statements acknowledging the significance of provincial boundaries, and that they did take them into consideration. For example, the 2007 report stated, “Either option would involve a breach of a county and a provincial boundary” and the 2002 report acknowledged that the existing Longford-Roscommon constituency breached a provincial boundary. In contrast, the reports from 2012 and 2017 did not mention provincial boundaries in a similar manner.

It should be noted that previous Commissions were not required to avoid breaching provincial boundaries. While past reports indicate that provincial boundaries have been considered in the formation of constituencies, they have not necessarily been weighted as heavily as other factors. For instance, in 2017, provincial boundaries were broken when an area in Offaly was added to the Tipperary constituency. Similarly, in 2012, part of Donegal was transferred to Sligo-Leitrim, and parts of Cavan were transferred to the same constituency. Additionally, in 2012, a portion of Tipperary was transferred to Offaly. Thus, we believe that breaching provincial boundaries should not preclude a potential constituency recommendation.

C.2 Variance Trade-off

The trade-off is choosing between equality of representation and respecting the terms of reference. This trade off looks like, from a mathematical perspective, increasing the acceptable variance up from $\pm 5\%$ to some higher threshold for inhomogeneity of representation to facilitate a lesser population transfers and disruption of county boundaries. The tighter the bound on the variance threshold, the less mathematical flexibility there is to adhere to terms of reference such as maintaining county boundaries. In this sense, the holistic view of representation aimed at in the terms of reference, and representation from a mathematical perspective are at odds.

An example of where this trade off might occur is with Laois and Offaly, see ?? for more details. In short, when the two counties are grouped together they should collectively be assigned a total of 6 seats. But at no point can the two counties independently form standalone 3 seat constituencies within the permitted $\pm 5\%$. A transfer of population and area must occur under the mathematical purview, the question is does strictly sticking to this perspective actually achieve greater representation? By contrast, does willingly departing from acceptable variance in order to have the counties form standalone constituencies, adhering more closely to the terms of reference, actually result in better representation in the holistic sense? [REFERENCE OUR SUBMISSION]

D Markov Chain Monte Carlo Methods

Monte Carlo methods are numerical algorithms involving random sampling used to calculate expected values. For identically distributed random variables $A_1, A_2, \dots$ with mean μ , then by the strong law of large numbers,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N A_i = \mu. \quad (24)$$

Thus, for a large enough N , the mean μ can be approximated by the sample mean

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N A_i. \quad (25)$$

A Markov chain is a sequence of “memoryless” states, i.e. states which are generated based on the previous state alone. More formally, for a (homogeneous) Markov chain of random variables $A_1, A_2, \dots$, then

$$\begin{aligned} P\{A_{N+1} = a_i \mid A_N = a_j, A_{N-1} = a_{j_{N-1}}, \dots, A_0 = a_{j_0}\} &= P\{A_{N+1} = a_i \mid A_N = a_j\} \\ &= P\{A_{M+1} = a_i \mid A_M = a_j\} \forall N, M. \end{aligned} \quad (26)$$

For a probability density function $\lambda(\sigma)$ that is to be sampled from using a Markov chain, define the transition probability density function

$$t(\sigma', \sigma) = P\{A_{N+1} = \sigma' \mid A_N = \sigma\} \quad (27)$$

and the infinitesimal volume element $d\Gamma_\sigma$ in the phase space centred around state σ . $\lambda(\sigma)$ and $t(\sigma', \sigma)$ are probability density functions, and so

$$\lambda(\sigma) \geq 0, \quad \int d\Gamma_\sigma \lambda(\sigma) = 1, \quad t(\sigma', \sigma) \geq 0, \quad \int d\Gamma_{\sigma'} t(\sigma', \sigma) = 1. \quad (28)$$

If $t(\sigma', \sigma)$ satisfies detailed balance for $\lambda(\sigma)$, that is

$$t(\sigma', \sigma)\lambda(\sigma) = t(\sigma, \sigma')\lambda(\sigma') \forall \sigma, \sigma', \quad (29)$$

then $\lambda(\sigma)$ is the equilibrium distribution of $t(\sigma', \sigma)$, i.e. choosing $t(\sigma', \sigma)$ as the transition distribution leads to a Markov chain of values sampled from $\lambda(\sigma)$ (as $\int d\Gamma_\sigma t(\sigma', \sigma)\lambda(\sigma) = \int d\Gamma_\sigma t(\sigma, \sigma')\lambda(\sigma') = \lambda(\sigma') \int d\Gamma_\sigma t(\sigma, \sigma') = \lambda(\sigma')$).

D.1 Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm (15) operates using an “accept/reject” process, outlined as follows (11).

Say $\lambda(\sigma)$ is a probability density function from which values are to be sampled. Let the proposal density function $p(\sigma', \sigma)$ be a distribution corresponding to proposing a state σ' given the current state σ , and the acceptance probability $a(\sigma', \sigma)$ be the probability that this proposed state is accepted as the next state, given by

$$a(\sigma', \sigma) = \min\left(\frac{p(\sigma, \sigma')\lambda(\sigma')}{p(\sigma', \sigma)\lambda(\sigma)}, 1\right). \quad (30)$$

These expressions satisfy

$$a(\sigma', \sigma)p(\sigma', \sigma)\lambda(\sigma) = a(\sigma, \sigma')p(\sigma, \sigma')\lambda(\sigma'), \quad (31)$$

and so if the transition probability density function is defined as

$$t(\sigma', \sigma) = a(\sigma', \sigma)p(\sigma', \sigma), \quad (32)$$

then $t(\sigma', \sigma)$ satisfies detailed balance for $\lambda(\sigma)$.

In the models proposed here, the proposed state σ' differs to the current state σ at a single lattice site i , i.e.

$$\sigma' = \sigma|_{q_i \rightarrow q'_i}, \quad (33)$$

and the algorithm is repeated for each site in turn.

The choice of the proposal function $p(\sigma', \sigma)$

D.1.1 Metropolis Algorithm

The Metropolis algorithm (16) is a specific case of the Metropolis-Hastings algorithm, where the proposal density function is symmetric about the current state, i.e. $p(\sigma', \sigma) = p(\sigma, \sigma')$. For $\lambda(\sigma)$ defined as in [Equation 13](#),

$$a(\sigma', \sigma) = \min\left(\frac{p(\sigma', \sigma)\lambda(\sigma')}{p(\sigma, \sigma')\lambda(\sigma)}, 1\right) = \min\left(\exp\left(\frac{-\Delta H(\sigma', \sigma)}{T}\right), 1\right), \quad (34)$$

and so only the difference $\Delta H(\sigma', \sigma) = H(\sigma') - H(\sigma)$ between the proposed and current Hamiltonian is necessary to calculate for producing the next state. Notably, if the proposed state reduces the Hamiltonian, then it is always accepted.

The simplest choice of a symmetric proposal function for updating site i according to [Equation 33](#) is a uniform distribution over the other states, i.e.

$$p(\sigma', \sigma) = \begin{cases} \frac{1}{Q-1}, & q'_i \neq q_i \\ 0, & q'_i = q_i \end{cases}. \quad (35)$$

D.1.2 Gibbs Sampler

The Gibbs sampler method (17) is another type of Metropolis-Hastings algorithm, where the proposed state is generated such that it is always accepted. In particular, the proposal density function is given by (11)

$$p(\sigma', \sigma) \propto \lambda(\sigma'), \quad (36)$$

and so

$$a(\sigma', \sigma) = \min\left(\frac{\lambda(\sigma)\lambda(\sigma')}{\lambda(\sigma')\lambda(\sigma)}, 1\right) = 1. \quad (37)$$

The proposal function $p(\sigma', \sigma)$ for single-site updating according to [Equation 33](#) has the same form as $\lambda(\sigma')$, differing only by a renormalising factor. For $\lambda(\sigma)$ defined as in [Equation 13](#),

$$p(\sigma', \sigma) = \frac{\lambda(\sigma')}{\sum_{\sigma''} \lambda(\sigma'')} = \frac{\exp\left(-\frac{H(\sigma')}{T}\right)}{\sum_{\sigma''} \exp\left(-\frac{H(\sigma'')}{T}\right)} = \frac{\exp\left(-\frac{\Delta H(\sigma', \sigma)}{T}\right)}{\sum_{\sigma''} \exp\left(-\frac{\Delta H(\sigma'', \sigma)}{T}\right)}. \quad (38)$$